🌀 M003 Extension — The Loop Pattern (Intermediate)

Topic: One loop, three counters (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Intermediate, after Week 6) · **Time:** about 40 minutes

One loop builds all **7** obsidian walls. It keeps three counters — **x**, **y**, **z** — and nudges each by a fixed **step** every pass. That's an **algorithm**. Type `SPIRAL` and watch.

```
x = 14
y = 0
z = 2
for step in range(7):
    fill OBSIDIAN from (14, 0, z) to (x, y, z)
    x = x - 2
    y = y + 1
    z = z + 2
```

> 🔁 **Big idea:** the `fill` line **never changes** — it is the same seven times. The walls come out **different** only because **x, y, z change** between passes.

Each pass the new wall is **2 wider** (x), **1 taller** (y), and **2 deeper** (z) than the last.

> 🔴 Stand on your **home spot** (red block) every run.
>
> 🎨 Every wall is **obsidian**.

1 · Predict 🔮

Trace the loop. Write the **START** values (x, y, z when the pass begins) and the **END** values (after the updates). Iteration 0 is done. **END of one row = START of the next.**

Iteration	START x	START y	START z	END x	END y	END z
0	14	0	2	12	1	4
1						
2						
3						
4						
5						
6						

The `fill` line runs *before* the updates. Which x, y, z does iteration 0's wall use? Circle one: (14, 0, 2) · (12, 1, 4)

Which counter makes each wall taller? Circle one: x · y · z

`range(7)` builds how many walls — and why does each wall come out bigger even though the `fill` line is never edited? (1–2 sentences)

2 · Spot the Bug 🐛

Each loop is broken. Read what it should do, fix it, and say why.

Bug A — a counter that never steps. The `z = z + 2` line was deleted, so z stays **2** the whole time.

```
for step in range(7):
    fill OBSIDIAN from (14, 0, z) to (x, y, z)
    x = x - 2
    y = y + 1
```

What do you see instead of a pattern? Circle one: all 7 walls stacked at z = 2 · 7 walls spread out

Write the missing line and where it goes:

Bug B — wrong step size. A friend changed one line to `z = z + 1`. It runs, but the 7 steps merged into one solid slope with no gaps between them.

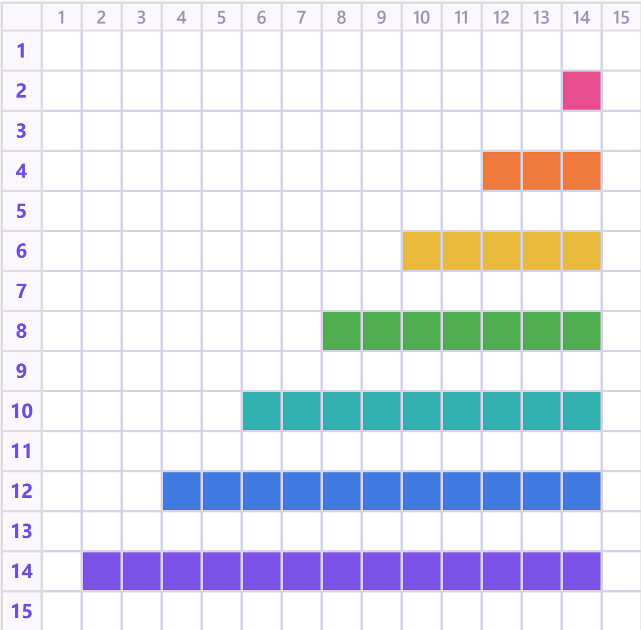
The walls are each 1 block thick. Explain why `z = z + 2` leaves a gap but `z = z + 1` makes the steps touch, then write the fix.

3 · Show Your Work 📸 🎥

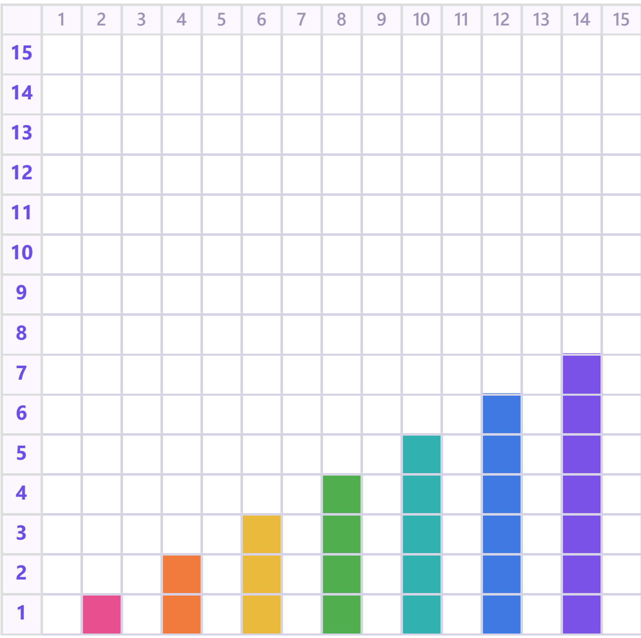
Load the world. Stand on your home spot. Type `SPIRAL` and watch all 7 walls build.

🧱 Same build, two views. Colour = the z value.

👁️ **Top view — looking DOWN (x → across, z ↓ deeper)**



↔️ **Side view (z → deeper, y ↑ up)**



colour = z value: 2 4 6 8 10 12 14

Each pass is 1 block taller — `y = y + 1` runs every loop.

Modify. Change **one** thing, predict, then build. E.g. `range(7)` → `range(10)`, or `y = y + 2`.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what the loop does and what each counter changes.

2 · Your build. Point the camera at the pattern. Name the smallest and biggest step.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your **one** video to teacher on KakaoTalk.